

30.5 A 321-Layer 2Tb 4b/cell 3D-NAND-Flash Memory with a 75MB/s Program Throughput

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The rising demand for high-density and high-performance NAND Flash is unprecedented, due to the expansion of data within enterprise SSDs, which fueled by the growth of the artificial intelligence industry. Both sequential- and random-access performance must improve to accommodate high-speed and large data transfers between a NAND Flash and its controller. Furthermore, improved reliability, and effective read and write performance continue to be crucial factors for QLC NAND Flash.

This paper introduces a 2Tb 321-layer 4-level cell (QLC) for a 3D NAND-Flash memory that utilizes a 6-plane architecture, with a 16kB page size, and 1.3ms page program time. Its bit density achieves 28.8Gb/mm² despite a 6-plane architecture. Implementation of a 6-plane structure reduces cost by employing a process like that of a 4-plane architecture; specifically, by utilizing the same metal layers. Furthermore, it is important to maintain a consistent internal voltage level within a chip, which is sourced from an external power V_{CC} pad. Reducing the V_{CC} supply resistance and preventing a significant increase in the area housing the charge pump is essential, even as the number of planes grouped into pages and blocks in a NAND die is increased. V_{CC} resistance increases with the size of the chip, but it is also constrained to fit within a limited package size, as shown in Fig. 30.5.1(a). To improve V_{CC} resistance, we use horizontal V_{CC} lines that are connected to vertical V_{CC} lines; yielding a V_{CC} resistance that is comparable to a 4-plane architecture, as shown in Fig. 30.5.1(b). To accommodate the space needed for six planes, the high-voltage regulators (HV REGs) are relocated from their initial location, next to the charge pumps, to the global WL decoders (GW DECs). The incorporation of highly conductive metal between the charge pump and the high-voltage regulator leads to a reduced resistance; thereby, helping to maintain resistance levels like those previously observed, as shown Fig. 30.5.1(c) and (d).

Improving program disturbance in the QLC device is crucial for ensuring wider read margins, despite the restrictions imposed by the narrow program/erase window and the sixteen program states. Figure 30.5.2(a) illustrates the influence of V_{CORE} , the internal cell supply voltage, on the NAND cell array channel. An increased V_{CORE} enhances the initial channel-charge level and reduces the boosted channel-leakage current; mitigating program disturbances without affecting the other 15 program states, as shown in Fig. 30.5.2(b). However, increasing V_{CORE} also increases current. To reduce average current and improve program disturbance, a V_{CORE} that only rises during the program-pulse phase is implemented in Fig. 30.5.2(c). As displayed in Fig. 30.5.2(d), this method leads to a 2% increase in average current, whereas the read level margin between erase and programmed cells shows an 18% improvement.

Peripheral circuits are required to exhibit a precise temperature coefficient (TC) for their bias voltages to eliminate temperature-dependent variations in a cell's V_{th} distribution. In this section, we introduce a bias generator that incorporates a controllable TC as depicted in Fig. 30.5.3(a). The bias generator comprises of 2 pairs of R_1 and R_2 resistor in series and a fully-differential operational-transconductance amplifier (OTA) with common-mode feedback (CMFB). The output voltages of the fully-differential OTA can be defined using the following equation:

$$OUTC, P = CTAT \pm R_2 / R_1 (CTAT - PTAT)$$

CTAT is the complementary-to-absolute-temperature coefficient, while PTAT is the proportional-to-absolute-temperature coefficient. By aligning CTAT and PTAT at a specific temperature we calibrate TC, since changing the feedback factor (R_2 / R_1) does not change the output bias. The procedure used is as follows; (1) CTAT and PTAT are adjusted to the same level at a given temperature; (2) the ratio of R_2 / R_1 is swept across different temperatures until the output bias reaches its target value. Figure 30.5.3(b) demonstrates the relationship between the feedback factor and the corresponding output bias. OUTC and OUTP are calibrated at a low temperature, and are still accurate at a high temperature, indicating that the output can be precisely controlled across a range of temperatures.

Power and signal integrity play a crucial role in determining the performance of interface circuits, and their importance increases with increasing data rates. A low-tap termination (LTT) is used to improve PI for low-power operation, to address limitation in spatial capacity for enhancing power-distribution network (PDN). Furthermore, center-tap termination (CTT) is used to guarantee interoperability with prior systems. To support both CTT and LTT interfaces, as depicted in Fig. 30.5.4(a), the NAND transmitter's pull-up driver is constructed using PMOS and NMOS transistors, while the pull-down driver is constructed using only NMOS transistors [1]. For high density, a stacked multi-die structure is preferred, rather than a single die, leading to an increase in the total parasitic IO-channel capacitance: due to the multitude of stacked dies and total IO capacitance (C_{IO}). To mitigate SI degradation, it is important to minimize C_{IO} . Improving transistor current drive reduces C_{IO} but also results in increased leakage. To mitigate leakage, a header is included in the NMOS of the pull-up driver. A large header is necessary for the pull-up PMOS to reduce V_{GS} deterioration resulting from the header V_{DS} drop. Hence, removing the pull-up PMOS header decreases the Tx area and minimizes C_{IO} , while adjusting the size of the pull-up PMOS results in leakage reduction. Removing the pull-up PMOS header also results in a 55% Tx area and a 20% C_{IO} reduction, as illustrated in Fig. 30.05.4(b). This reduces C_{IO} to meet the high-speed data requirements, allowing for 3.2Gb/s data, while being compatible with a range of form factors.

A die photo along with a key-parameter comparison table in relation to previous work is shown in Fig. 30.5.5 and 30.5.6 [2,3].

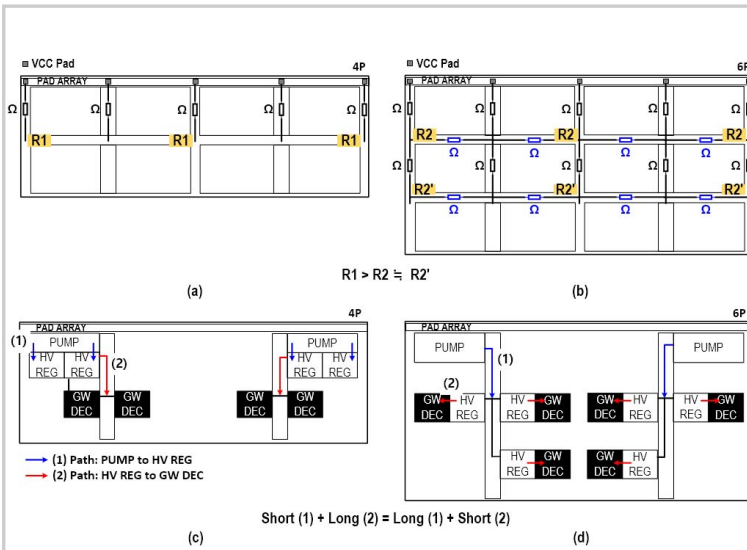


Figure 30.5.1: Comparison of 4-plane versus 6-plane placement and routing on reducing V_{CC} resistance and line resistance along the path from the charge pump to the global-WL decoder.

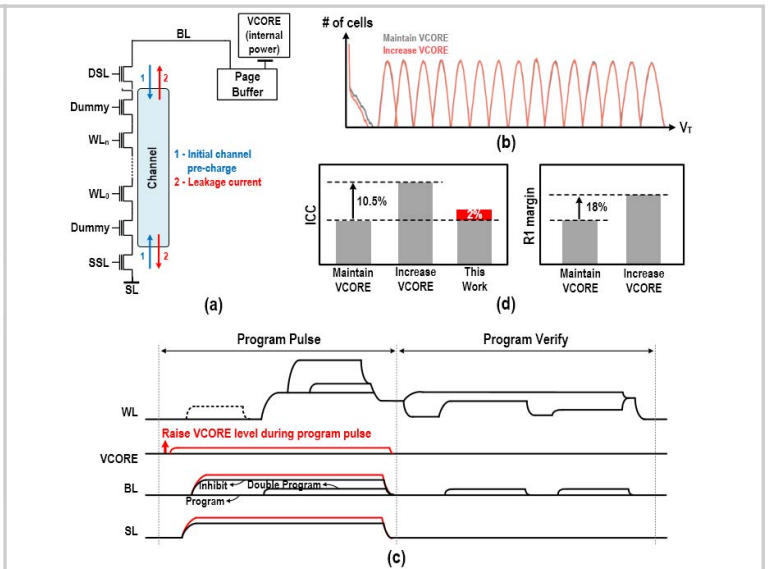


Figure 30.5.2: (a) relationship between V_{CORE} and channel, (b) V_{CORE} -level dependent program disturbance (c) V_{CORE} rise during program pulse, and (d) resulting ICC and $R1$ margins.

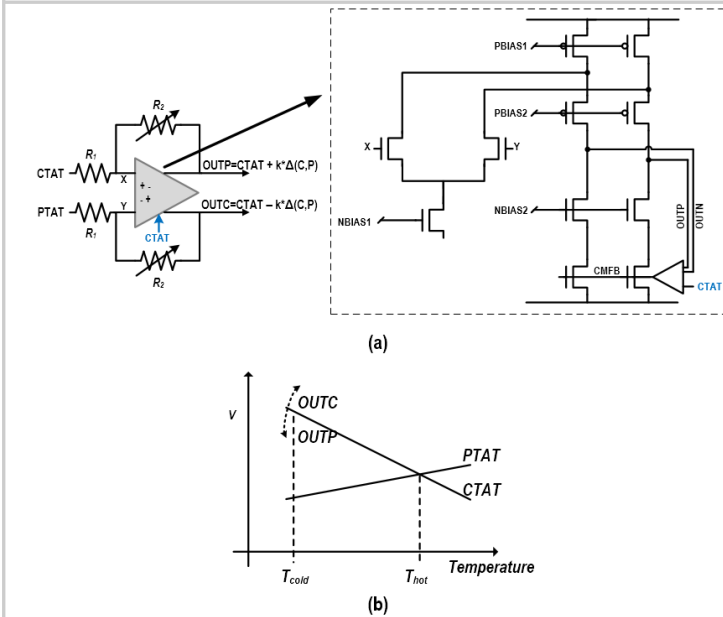


Figure 30.5.3: (a) Proposed bias generator and (b) its output vs temperature.

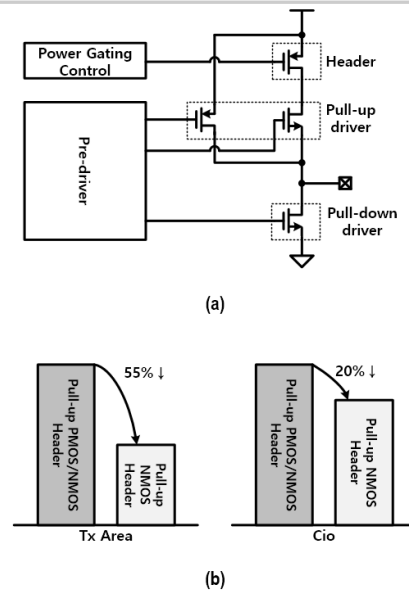


Figure 30.5.4: (a) Proposed Tx-interface scheme and (b) its circuit are and C_{io} improvement.

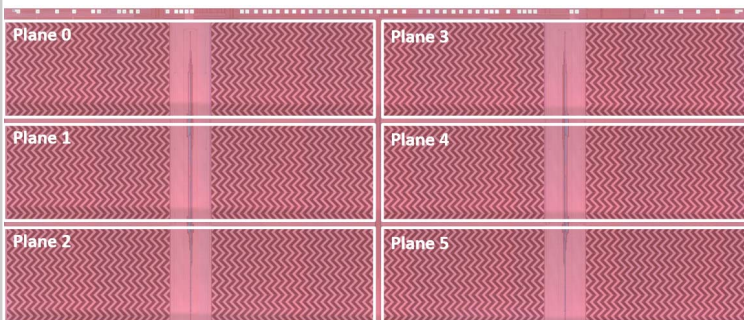


Figure 30.5.5: Die photograph of the 2Tb 4b/cell 321-layer 3D NAND Flash using a 6-plane architecture.

	This Work	ISSCC '24	ISSCC '22
Bits per cell	4	4	4
Density	2Tb	1Tb	1Tb
Plane	6	4	4
Technology	321-WL-Layer 3D NAND Flash Memory	280-WL-Layer 3D NAND Flash Memory	176-WL-Layer 3D NAND Flash Memory
Bit Density	28.8Gb/mm ²	28.5Gb/mm ²	14.8Gb/mm ²
Program Throughput	75MB/s	41MB/s	40MB/s
tR	80us	85us	90us
IO Speed	3200MB/s	3200MB/s	1600MB/s

Figure 30.5.6: Key parameter comparison table.

References:
[1] J. Cho, et al., "A 512Gb 3b/Cell 7th -Generation 3D-NAND Flash Memory with 184MB/s Write Throughput and 2.0Gb/s Interface," *ISSCC*, pp. 426-428, 2021.
[2] W. Jung et al., "A 280-Layer 1Tb 4b/cell 3D-NAND Flash Memory with a 28.5Gb/mm² Areal Density and a 3.2GB/s High-Speed IO Rate," *ISSCC*, pp. 236-237, 2024.
[3] W. Cho et al., "A 1-Tb, 4b/Cell, 176-Stacked-WL 3D-NAND Flash Memory with Improved Read Latency and a 14.8Gb/mm² Density," *ISSCC*, pp. 134-135, 2022.